/*
COP3502C | Fall 2025 | Section 0004
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Mission 1: zeros(n)

I count trailing zeros in n! by adding how many 5s are in all the factors. The recursive rule in my zeros function is: if n == 0, return 0; otherwise return n/5 + zeros(n/5). That naturally adds 5s from 25, 125, etc. Example I tested: zeros(100) = 24.

Mission 2: isPalindrome(str, left, right)

I check the ends and move inward. If left >= right, it's a palindrome and I return 1. If str[left] != str[right], I return 0. Otherwise I call isPalindrome on (left+1, right-1). With my tests, "racecar" returns 1 and "hello" returns 0.

Mission 3: findMax(arr, start, end) and findMin(arr, start, end)

I use a closed interval [start, end]. If start == end, I just return that element. For max, I compare arr[start] with the result of findMax on the rest (start+1 to end) and return the bigger one. For min, I do the same but return the smaller one. On my test array {5, 2, 9, 1, 7}, my functions gave max = 9 and min = 1.

Mission 4: searchRecord(numbers, low, high, value)

This is a recursive binary search with inclusive bounds. If low > high, I return -1 because it's not found. I set mid = low + (high - low) / 2. If numbers[mid] == value, I return mid. If numbers[mid] > value, I search the left side by calling searchRecord(numbers, low, mid - 1, value). Otherwise I search the right side by calling searchRecord(numbers, mid + 1, high, value). I tested on [1, 3, 5, 7, 9, 11]. Searching 7 returned index 3. Searching 8 returned -1.